DATA ITEM DESCRIPTION

Form Approved
OMB No. 0704-0188

Public reporting burden for this collection of information is estimated to average 110 hours per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden, to Washington Headquarters Services, Directorate for Information Operations and Reports, 1215 Jefferson Davis Highway, Suite 1204, Arlington, VA 22202-4302, and to the Office of Management and Budget, Paperwork Reduction Project (0704-0188), Washington, DC 20503.

1. TITLE	2. IDENTIFICATION NUMBER
Functional Flow Diagrams	DI-GDRQ-81224

3. DESCRIPTION/PURPOSE

3.1 The Functional Flow Diagrams are a portrayal of the functions which shall be met to satisfy total system needs and thus represents the beginning of subsequent detail requirements determination.

(Continued on Page 2)

4. APPROVAL DATE (YYMMDD)	5. OFFICE OF PRIMARY RESPONSIBILITY (OPR)	6a. DTIC APPLICABLE	6b. GIDEP APPLICABLE
910626	A/MICOM		

7. APPLICATION/INTERRELATIONSHIP

7.1 This Data Item Description (DID) contains the format and content preparation instructions for the data product generated by the specific and discrete requirement as delineated in the contract.

7.2 This DID should be used when a multi-contractor relationship exists and there is a need for commonality of this data among the contractors.

(Continued on Page 2)

8. APPROVAL LIMITATION	9a. APPLICABLE FORMS	9b. AMSC NUMBER
		A6659

10. PREPARATION INSTRUCTIONS

10.1 Content and format. The Functional Flow Diagrams content and format shall be as follows.

10.1.1 Functioning numbering. Functions on top-level functional diagram shall be numbered 1.0, 2.0, 3.0, etc. Functions which further indenture these top functions shall contain the same parent identifier and shall be coded at the next decimal level for each indenture. For example, the first indentured function 3.0 would be 3.1, the second 3.1.1, the third 3.1.1.1, etc. In expansion of a higher level function within a particular level of indenture, a numerical sequence shall be used to preserve the continuity of function. For example, if more than one function is required to amplify the function 3.0 at the first level of indenture, the sequence shall be 3.1, 3.2, 3.3, ...3.n. In expansion of function 3.3 at the second level, the numbering shall be 3.3.1, 3.3.2, 3.3.n. Where several levels of indentures appear on a single functional diagram, the same pattern shall be maintained. While the basic ground rule shall be to maintain a minimum level of indenture of any one particular flow, it may become necessary to include several levels to preserve the continuity of functions and to minimize the

(Continued on Page 2)

11. DISTRIBUTION STATEMENT

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

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Block 3. Description/Purpose (Continued)

3.2 These diagrams are drawings to be approved and released internally within a contractor's facility and are used as an integral part of the system engineering documentation process. The functional description is also an aid in developing, interpreting, and providing standardization necessary to accomplish interfaces between contractors.

Block 7. Application/Interrelationship (Continued)

7.3 This DID supersedes DI-S-3604.

Block 10. Preparation Instructions (Continued)

number of flows required to functionally depict the system. The general criteria for the number of functions and level of indentures appearing on any particular flow shall be accuracy and clarity of presentation based on judgement.

10.1.2 Function reference. Each functional diagram shall contain a reference to its next higher functional diagram through the use of a reference block. For example, function 4.3 should be shown as a reference block in the case where the functions 4.3.1, 4.3.2, 4.3.n, are being used to expand function 4.3; reference blocks shall also be used to indicate interfacing functions as appropriate.

10.1.3 Function box. Each separate function on a functional diagram shall be presented in a single box enclosed by a solid line as depicted in figure 1. Boxes used for reference to other flows shall be indicated as partially enclosed boxed to be labeled "Ref." Each function may be as gross or detailed as required by the level of functional diagram on which it appears, but it shall stand for a definite, finite, discrete action to be accomplished by equipment, personnel, facilities, or any combination of the three. Questionable or tentative functions shall be enclosed in dotted boxes as depicted by figure 1.

10.1.4 Flow connection. Lines connecting functions shall indicate only the functional flow and shall not represent either a lapse in time or any intermediate activity. In indicating the flow, vertical and horizontal lines between boxes shall indicate that all functions so interrelated must be performed in either a parallel or series sequence. Diagonal lines may be used to indicate alternate sequences (cases where alternate paths lead to the next function in the sequence). In this latter case, the use of a diagonal line indicates that any of the functions so interrelated shall lead to the next indicated function.

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10.1.5 Flow direction. Functional diagrams shall be laid out so that the functional flow is from left to right and the reverse flow, in the case of a functional loop, from right to left. Primary input lines shall enter the function box from the left side; the primary output or "go" line shall exist from the right and the "no go" line from the bottom of the box. However, where other considerations dictate a different arrangement to highlight a physical area, level of maintenance, or other significant consideration, a different arrangement may be employed.

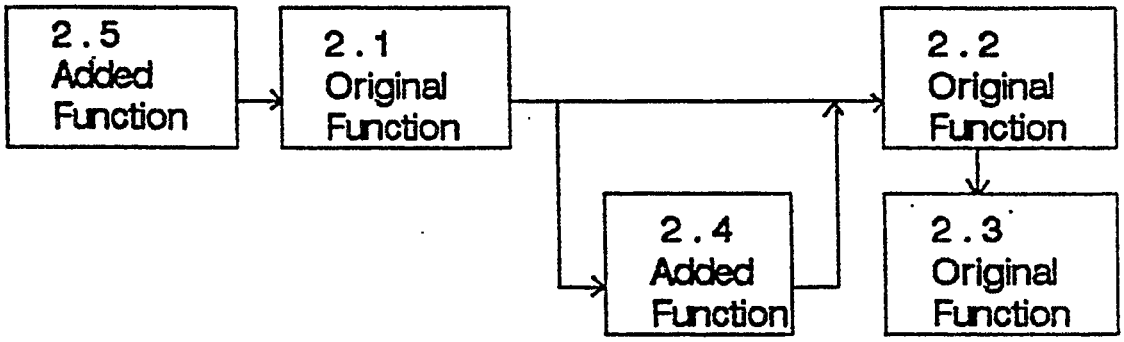
10.1.6 Summing gates. A circle shall be used to depict a summing gate. As in the case of functional boxes, lines shall enter or exit the summing gate as appropriate (reference figure 1). The summing gate is used to indicate the convergence or divergence of parallel or alternate functional paths and is annotated with the terms "AND" or "OR" respectively. The term "AND" is used to indicate that parallel functions leading into the gate must be accomplished before proceeding into the next function or that paths emerging from the "AND" gate must be accomplished after the preceding function. The term "OR" is used to indicate that any of several alternate paths (alternate functions) converge to or diverge from the "OR" gate. The "OR" gate thus indicates that alternate paths may lead to or follow a particular function.

10.1.7 GO/NO paths. The symbols "G" and "G" are used to indicate "GO" and "NO GO" paths. The symbols are entered adjacent to the lines leaving a particular function to indicate alternate functional paths. (Reference function 4.3.7).

10.1.8 Numbering procedures for changes to functional diagrams. In order to provide a rapid means of changing flows without causing extensive or chain reaction revision of numbering, the following procedure shall be used. Additions of functions to existing data shall be accomplished by locating the new function in its correct position without regard to sequence of numbering. The new function shall be numbered using the first unused number at the level of indenture appropriate for the new function.

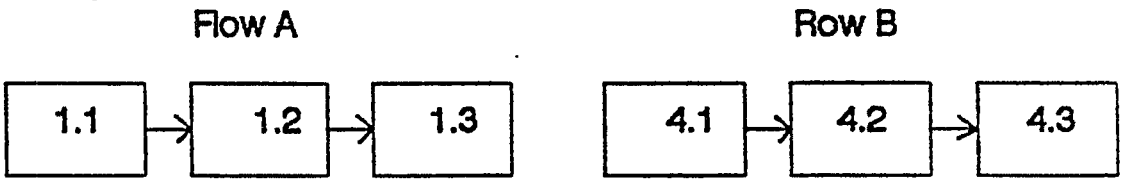
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Example:

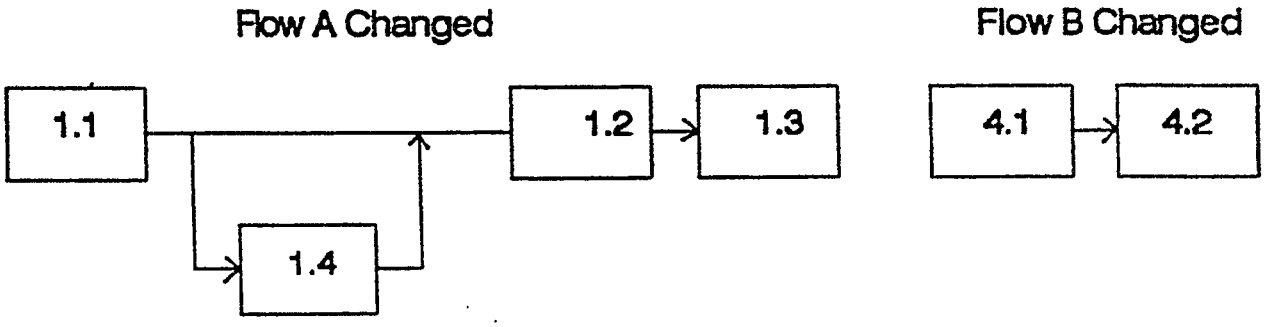


When previously established functions must be redelegated to a different functional string, the function to be moved shall be considered retired and the new location of that function shall be considered as a new addition to the acquiring strings and shall be treated as above.

Example:



Change: Function 4.3 from flow B to flow A



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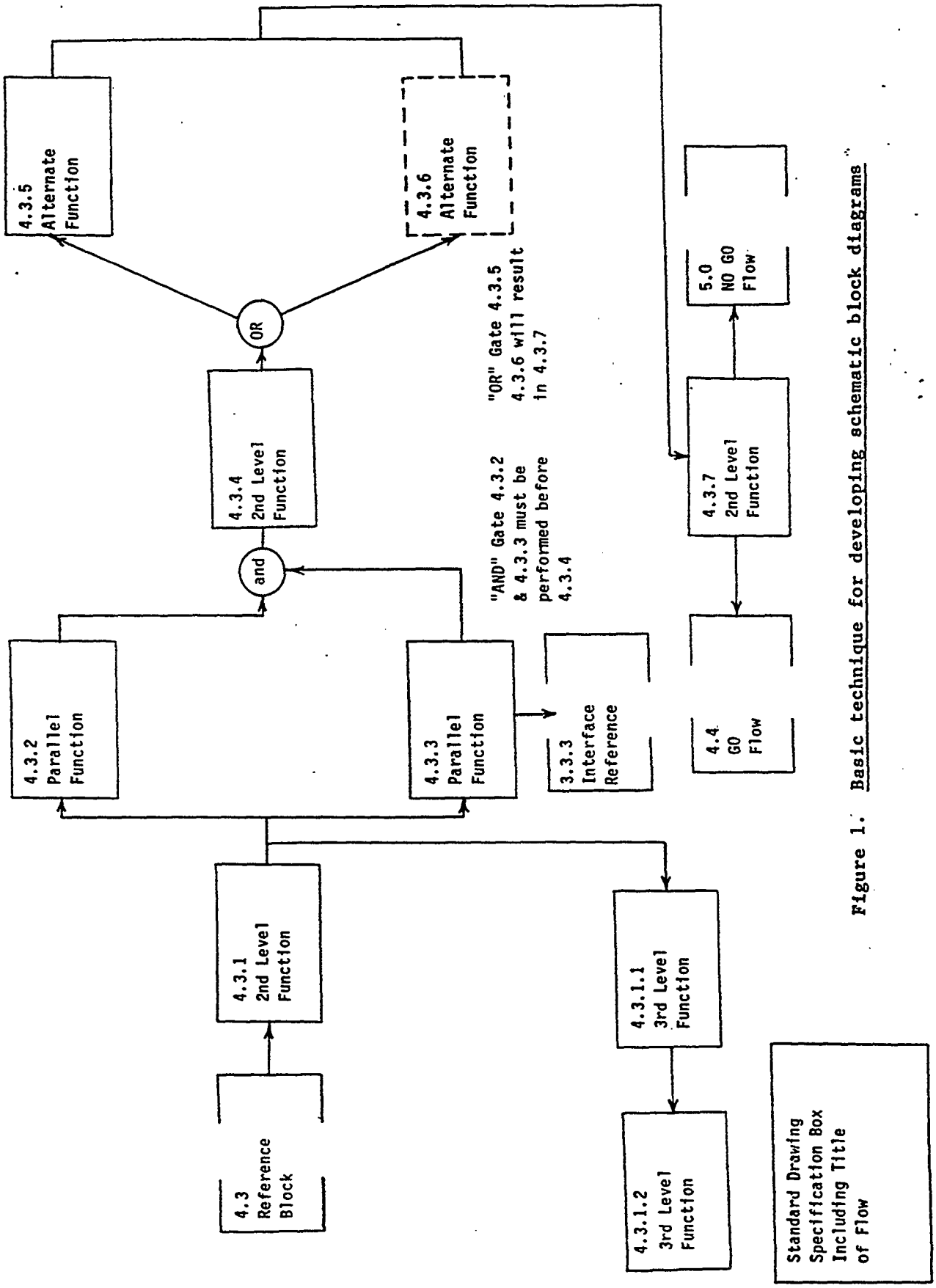


Figure 1. Basic technique for developing schematic block diagrams

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